



Department of Management Sciences

Course: Databases for Business Analytics

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Lab Final Report

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Database Design Final Project

J. Alam Business Database

ERD, Normalization, Relational Schemas, Codes

1. EXECUTIVE SUMMARY & BUSINESS SCOPE:

1.1 About the Business

J.Alam is a fashion resale business that operates natively through Facebook Live sessions. Inventory is acquired from major Pakistani fashion retailers (such as *Sapphire*, *Khaadi*, and *Generation*) during discounted clearance cycles. Items are managed from a residential inventory space and sold to clients at competitive, sub-retail rates (~20% lower than standard boutique prices).

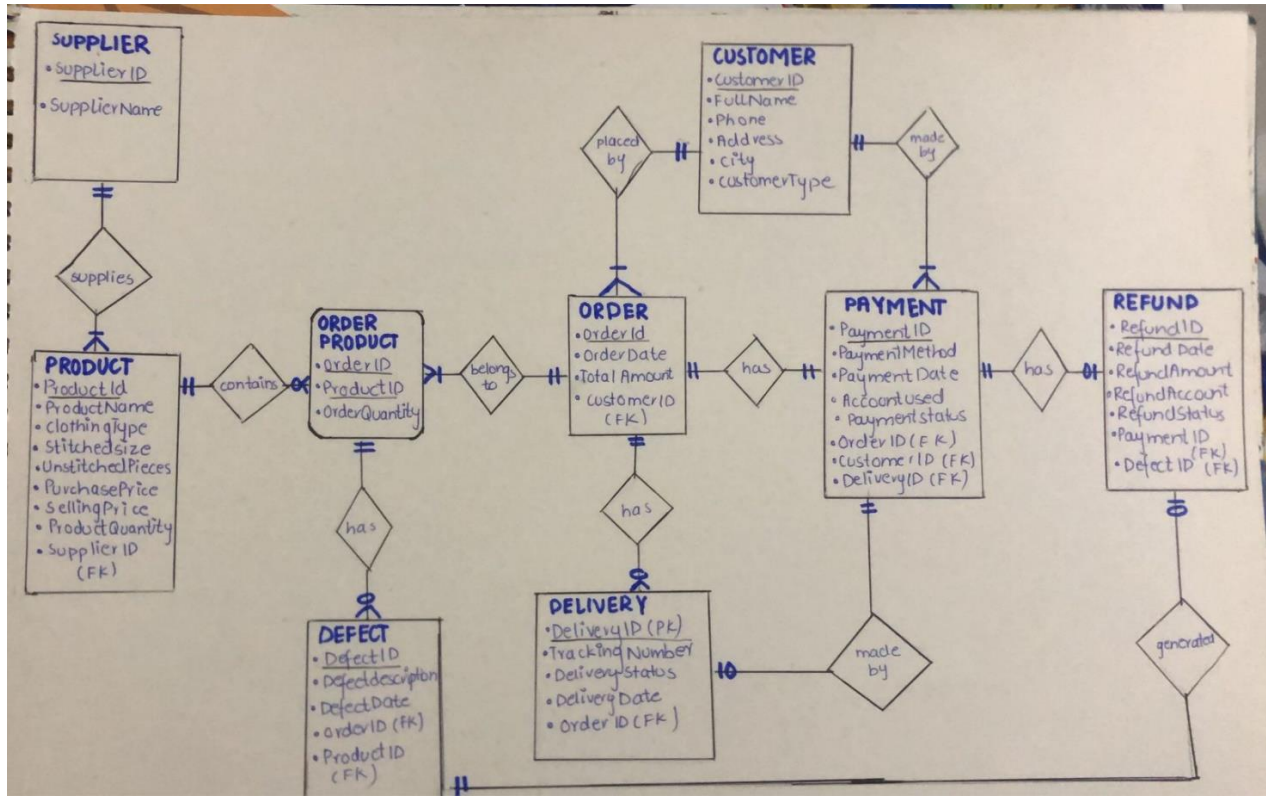
1.2 Operational Challenges & Database Purpose

Transactions originate on Facebook Live and migrate to WhatsApp for customer confirmations. Order completion relies either on home pickup or third-party logistics via TCS courier services.

The primary business problem stems from manual record-keeping on paper slips, causing data gaps in sales volumes, net profit performance, customer purchase history, and logistics monitoring. This relational database transitions J.Alam to a digitized architecture designed to:

- Quantify exact customer orders, item variations, and specific quantities.
- Track transactions across distinct fulfillment workflows (Advance payments, Cash-on-Delivery, and Cash-in-Hand) across multiple designated bank tracking channels.
- Verify live shipping statuses using tracking keys integrated with TCS delivery workflows.
- Isolate loyal buyers based on historic transaction trends to evaluate COD clearings.
- Dynamically compute profit configurations per SKU (Selling Price minus Purchase Cost).

2. Entity-Relationship Diagram:



3. Normalization:

Normalization is the process of organizing database tables to remove data redundancy and ensure data is stored logically and consistently. Three levels are used in this project:

- **1NF:** Every cell must contain one single value. No repeating groups. Table must have a Primary Key.
- **2NF:** Must be in 1NF. Every non-key attribute must depend on the FULL Primary Key. Partial dependency is only checked when a composite Primary Key exists.
- **3NF:** Must be in 2NF. No non-key attribute should depend on another non-key attribute. No transitive dependency allowed.

Tables already in 3NF are shown with full data and a written reason. Tables that violate 3NF are restructured with new tables created to remove the violation.

3.1 Unnormalized Forms to 3NF Transformations

Table 1: Supplier (Strong Entity)

SupplierID (PK)	SupplierName
S01	Sapphire
S02	Khaadi
S03	Generation
S04	Ethnic by Outfitters

Interpretation:

The Supplier entity stores information about the brands or businesses supplying products to the store. Each supplier has a unique SupplierID which identifies the supplier record. All attributes directly depend on the primary key, and there are no repeating groups or transitive dependencies.

CONCLUSION:

Supplier is already in 3NF. No normalization required.

Table 2: Customer (Strong Entity)

CustomerID (PK)	FullName	Phone	Address	City	CustomerType
C01	Ayesha Tariq	03001234567	House 12 Block B Gulberg III	Lahore	Old
C02	Fatima Malik	03111234568	Flat 5 Clifton Block 4	Karachi	Old
C03	Sara Hussain	03211234569	House 7 F-8 Markaz	Islamabad	Old
C04	Zara Ahmed	03311234560	House 3 Hayatabad Phase 2	Peshawar	New
C05	Hina Baig	03421234561	Flat 9 Bahria Town Phase 4	Rawalpindi	Old
C06	Sana Qureshi	03501234562	House 22 DHA Phase 6	Lahore	Old
C07	Maria Iqbal	03021234563	Flat 2 North Nazimabad Block H	Karachi	New
C08	Nadia Raza	03121234564	House 45 G-10 Markaz	Islamabad	Old
C09	Asma Khan	03221234565	House 8 Model Town	Lahore	New
C10	Rabia Ali	03321234566	Flat 11 Defence Phase 2	Karachi	Old
C11	Ume Kulsoom	03401234570	House 6 Johar Town	Lahore	New
C12	Bushra Noor	03451234571	Flat 14 Gulshan-e- Iqbal Block 3	Karachi	Old
C13	Amna Siddiqui	03061234572	House 18 E-7 Islamabad	Islamabad	New

C14	Rida Farooq	03161234573	House 9 Saddar Cantonment	Rawalpindi	Old
C15	Maham Riaz	03261234574	Flat 3 Bahria Town Precinct 12	Karachi	New
C16	Iqra Shahid	03361234575	House 33 Cavalry Ground	Lahore	Old
C17	Nimra Javed	03461234576	House 5 Satellite Town	Rawalpindi	New
C18	Kiran Nasir	03071234577	Flat 7 F-11 Markaz	Islamabad	Old
C19	Sadia Pervaiz	03171234578	House 2 Wapda Town Phase 1	Lahore	New
C20	Maryam Zahid	03271234579	Flat 6 Block 13 Gulshan	Karachi	Old

Interpretation:

The Customer entity contains customer details such as name, phone number, address, city, and customer type. CustomerID uniquely identifies each customer and all other attributes depend directly on it. No attribute depends on another non-key attribute, so there is no transitive dependency in the table.

CONCLUSION:

Customer is already in 3NF. No normalization required.

Table 3: Product (Strong Entity)—> Normalization Required

Note on ProductQuantity Attribute:

ProductQuantity is the quantity of each product that J.Alam currently has in her inventory at home. This is different from OrderQuantity in OrderProduct which is how many units a customer ordered. ProductQuantity tells us how many pieces of each product are available for sale.

Original Product Table Before Normalization

ProductID	Product Name	ClothingType	StitchSize	UnstitchPieces	PurchasePrice	Selling Price	ProductQuantity	PurchaseDate	SupplierID
P01	Sapphire Embroidered 3PC Lawn Suit	Unstitched	NULL	3 Piece	5500	7000	8	12-Feb-2024	S01
P02	Sapphire Printed 2PC Lawn	Unstitched	NULL	2 Piece	3800	4800	12	12-Feb-2024	S01
P03	Sapphire Lawn Stitched Shirt	Stitched	10	NULL	2800	3500	15	15-Feb-2024	S01
P04	Sapphire Khaddar Stitched Kurta	Stitched	12	NULL	2400	3000	10	15-Feb-2024	S01
P05	Sapphire Winter Embroidered 1PC	Unstitched	NULL	1 Piece	2200	2800	20	20-Feb-2024	S01
P06	Khaadi Embroidered 3PC Khaddar	Unstitched	NULL	3 Piece	6000	7800	6	03-Mar-2024	S02
P07	Khaadi Printed	Unstitched	NULL	2 Piece	4000	5200	14	03-Mar-2024	S02

	2PC Lawn								
P08	Khaadi Khaddar Stitched Kurta	Stitched	14	NULL	3200	4000	9	05-Mar- 2024	S02
P09	Khaadi Stitched Shirt Size 8	Stitched	8	NULL	2600	3300	11	05-Mar- 2024	S02
P10	Khaadi Printed 1PC Dupatta	Unstitch ed	NULL	1 Piece	1500	2000	25	10-Mar- 2024	S02
P11	Generati on Embroid ered 3PC	Unstitch ed	NULL	3 Piece	4800	6200	7	18-Mar- 2024	S03
P12	Generati on Printed 1PC Dupatta	Unstitch ed	NULL	1 Piece	1800	2400	18	18-Mar- 2024	S03
P13	Generati on Stitched Linen Shirt	Stitched	16	NULL	2500	3200	13	20-Mar- 2024	S03
P14	Generati on Lawn 2PC Suit	Unstitch ed	NULL	2 Piece	3500	4500	9	22-Mar- 2024	S03
P15	Generati on Stitched Formal Kurta	Stitched	10	NULL	3000	3800	8	25-Mar- 2024	S03

P16	Ethnic Printed Lawn 3PC	Unstitched	NULL	3 Piece	3800	4800	10	10-Apr-2024	S04
P17	Ethnic Embroidered 2PC	Unstitched	NULL	2 Piece	3500	4500	12	10-Apr-2024	S04
P18	Ethnic Stitched Casual Kurta	Stitched	10	NULL	2000	2600	16	12-Apr-2024	S04
P19	Ethnic Printed 1PC Shirt	Unstitched	NULL	1 Piece	1600	2100	22	12-Apr-2024	S04
P20	Ethnic Stitched Khaddar Kurta	Stitched	12	NULL	2800	3500	7	15-Apr-2024	S04
P21	Sapphire Summer Printed 2PC	Unstitched	NULL	2 Piece	4200	5400	11	18-Apr-2024	S01
P22	Khaadi Summer Stitched Shirt	Stitched	14	NULL	2700	3400	9	20-Apr-2024	S02

Interpretation:

1NF: Every cell has one value. ProductID is the Primary Key. No repeating groups. PASS.

2NF: Single-column Primary Key. All attributes depend fully on ProductID. PASS.

3NF Violation Found

A transitive dependency exists in the original Product table:

- ClothingType is a non-key attribute.

- StitchedSize depends on ClothingType. When ClothingType is Stitched, StitchedSize has a value such as 8, 10, 12, 14 or 16. When ClothingType is Unstitched, StitchedSize is NULL.
- UnstitchedPieces also depends on ClothingType. When ClothingType is Unstitched, UnstitchedPieces has a value of 1 Piece, 2 Piece or 3 Piece. When ClothingType is Stitched, UnstitchedPieces is NULL.

Transitive dependency chain:

- ProductID -> ClothingType -> StitchedSize
- ProductID -> ClothingType -> UnstitchedPieces

StitchedSize and UnstitchedPieces do not directly depend on ProductID. They depend on ClothingType first. ClothingType is a non-key attribute determining other non-key attributes. This violates 3NF.

Two New Tables Created

Two separate tables are created instead of one combined table because stitched and unstitched attributes are completely different from each other:

- **ProductStitched:** stores ProductID and StitchedSize for stitched products only. ProductName is not included because it already exists in the main Product table and repeating it here would be redundant. For any stitched ProductID we can directly look up its size.
- **ProductUnstitched:** stores ProductID and UnstitchedPieces for unstitched products only. ProductName is not included for the same reason. For any unstitched ProductID we can directly look up its piece count.

StitchedSize and UnstitchedPieces are removed from the Product table. ClothingType remains in Product as it is a direct independent fact about the product. ProductQuantity also remains in Product as it is a direct fact about how many units are in inventory.

New Table: ProductStitched (9 rows —> stitched products only)

ProductID (PK)(FK)	StitchedSize
P03	10
P04	12

P08	14
P09	8
P13	16
P15	10
P18	10
P20	12
P22	14

New Table: ProductUnstitched (13 rows —> unstitched products only)

ProductID (PK)(FK)	UnstitchedPieces
P01	3 Piece
P02	2 Piece
P05	1 Piece
P06	3 Piece
P07	2 Piece
P10	1 Piece
P11	3 Piece
P12	1 Piece
P14	2 Piece
P16	3 Piece
P17	2 Piece
P19	1 Piece
P21	2 Piece

Updated Product Table After Normalization (22 rows)

ProductID	ProductName	ClothingType	PurchasePrice	SellingPrice	ProductQuantity	PurchaseDate	SupplierID (FK)
P01	Sapphire Embroidered 3PC Lawn Suit	Unstitched	5500	7000	8	12-Feb-2024	S01
P02	Sapphire Printed 2PC Lawn	Unstitched	3800	4800	12	12-Feb-2024	S01
P03	Sapphire Lawn Stitched Shirt	Stitched	2800	3500	15	15-Feb-2024	S01
P04	Sapphire Khaddar Stitched Kurta	Stitched	2400	3000	10	15-Feb-2024	S01
P05	Sapphire Winter Embroidered 1PC	Unstitched	2200	2800	20	20-Feb-2024	S01
P06	Khaadi Embroidered 3PC Khaddar	Unstitched	6000	7800	6	03-Mar-2024	S02
P07	Khaadi Printed 2PC Lawn	Unstitched	4000	5200	14	03-Mar-2024	S02
P08	Khaadi Khaddar Stitched Kurta	Stitched	3200	4000	9	05-Mar-2024	S02
P09	Khaadi Stitched Shirt Size 8	Stitched	2600	3300	11	05-Mar-2024	S02

P10	Khaadi Printed 1PC Dupatta	Unstitched	1500	2000	25	10-Mar-2024	S02
P11	Generation Embroidered 3PC	Unstitched	4800	6200	7	18-Mar-2024	S03
P12	Generation Printed 1PC Dupatta	Unstitched	1800	2400	18	18-Mar-2024	S03
P13	Generation Stitched Linen Shirt	Stitched	2500	3200	13	20-Mar-2024	S03
P14	Generation Lawn 2PC Suit	Unstitched	3500	4500	9	22-Mar-2024	S03
P15	Generation Stitched Formal Kurta	Stitched	3000	3800	8	25-Mar-2024	S03
P16	Ethnic Printed Lawn 3PC	Unstitched	3800	4800	10	10-Apr-2024	S04
P17	Ethnic Embroidered 2PC	Unstitched	3500	4500	12	10-Apr-2024	S04
P18	Ethnic Stitched Casual Kurta	Stitched	2000	2600	16	12-Apr-2024	S04
P19	Ethnic Printed 1PC Shirt	Unstitched	1600	2100	22	12-Apr-2024	S04
P20	Ethnic Stitched Khaddar Kurta	Stitched	2800	3500	7	15-Apr-2024	S04

P21	Sapphire Summer Printed 2PC	Unstitched	4200	5400	11	18-Apr-2024	S01
P22	Khaadi Summer Stitched Shirt	Stitched	2700	3400	9	20-Apr-2024	S02

CONCLUSION:

Product table violated 3NF because StitchedSize and UnstitchedPieces were transitively dependent on ClothingType. Fixed by creating ProductStitched and ProductUnstitched tables. These tables store only ProductID and their respective attribute — ProductName is not repeated as it already exists in the main Product table. ProductQuantity attribute correctly remains in Product as it directly describes each product's inventory quantity.

Table 4: Order (Strong Entity)

OrderID (PK)	OrderDate	TotalAmount (PKR)	CustomerID (FK)
O01	03-May-2024	3500	C01
O02	03-May-2024	3000	C03
O03	03-May-2024	9200	C05
O04	03-May-2024	6500	C08
O05	03-May-2024	3500	C12
O06	03-May-2024	7800	C16
O07	03-May-2024	5600	C18
O08	08-May-2024	3000	C02
O09	08-May-2024	3500	C06
O10	08-May-2024	7300	C09
O11	08-May-2024	13000	C10
O12	08-May-2024	3200	C14
O13	08-May-2024	2600	C17
O14	13-May-2024	8000	C01
O15	13-May-2024	13500	C03
O16	13-May-2024	5200	C04
O17	13-May-2024	3300	C07
O18	13-May-2024	4800	C11
O19	13-May-2024	3200	C15
O20	13-May-2024	6200	C20
O21	19-May-2024	7800	C02
O22	19-May-2024	3500	C05

O23	19-May-2024	8000	C08
O24	19-May-2024	2400	C13
O25	19-May-2024	4500	C19
O26	25-May-2024	6200	C06
O27	25-May-2024	5200	C10
O28	25-May-2024	7800	C12
O29	25-May-2024	3500	C16
O30	25-May-2024	3200	C18

Interpretation:

The Order entity stores customer order details including order date, total amount, and the customer who placed the order. OrderID uniquely identifies each order and all non-key attributes depend directly on it. No partial or transitive dependency exists within the table.

CONCLUSION:

Order is already in 3NF. No normalization required.

Table 5: OrderProduct (Associative Entity)

OrderID (PK)(FK)	ProductID (PK)(FK)	OrderQuantity
O01	P03	1
O02	P04	1
O03	P08	1
O03	P07	1
O04	P09	1
O04	P13	1
O05	P03	1
O06	P06	1
O07	P18	1
O07	P04	1
O08	P04	1
O09	P03	1
O10	P08	1
O10	P09	1
O11	P06	1
O11	P07	1
O12	P13	1
O13	P18	1
O14	P08	2
O15	P03	3
O15	P04	1

O16	P07	1
O17	P09	1
O18	P16	1
O19	P13	1
O20	P11	1
O21	P06	1
O22	P03	1
O23	P08	2
O24	P12	1
O25	P17	1
O26	P11	1
O27	P18	2
O28	P06	1
O29	P03	1
O30	P13	1

Interpretation:

The OrderProduct entity acts as an associative table between Order and Product entities. It uses a composite primary key consisting of OrderID and ProductID to uniquely identify each ordered product. The OrderQuantity attribute depends on both keys together, meaning there is no partial dependency.

CONCLUSION:

OrderProduct is already in 3NF. No normalization required.

Table 6: Payment (Weak Entity) —> Normalization Required**Original Payment Table Before Normalization (30 rows)**

PaymentID	PaymentMethod	PaymentDate	AccountUsed	PaymentStatus	OrderID	CustomerID	DeliveryID
PM01	COD	10-May-2024	Account B	Received	O01	C01	D01
PM02	COD	10-May-2024	Account B	Received	O02	C03	D02
PM03	COD	10-May-2024	Account B	Received	O03	C05	D03
PM04	COD	10-May-2024	Account B	Received	O04	C08	D04
PM05	Advance	03-May-2024	Account A	Received	O05	C12	NULL
PM06	COD	10-May-2024	Account B	Received	O06	C16	D05
PM07	COD	10-May-2024	Account B	Received	O07	C18	D06
PM08	COD	15-May-2024	Account B	Received	O08	C02	D07
PM09	COD	15-May-2024	Account B	Received	O09	C06	D08
PM10	Advance	08-May-2024	Account A	Received	O10	C09	NULL
PM11	COD	15-May-2024	Account B	Received	O11	C10	D09
PM12	COD	15-May-2024	Account B	Received	O12	C14	D10
PM13	Advance	08-May-2024	Account A	Received	O13	C17	NULL

PM14	COD	20-May-2024	Account B	Received	O14	C01	D11
PM15	COD	20-May-2024	Account B	Received	O15	C03	D12
PM16	Advance	13-May-2024	Account A	Received	O16	C04	NULL
PM17	Advance	13-May-2024	Account A	Received	O17	C07	NULL
PM18	COD	20-May-2024	Account B	Pending	O18	C11	D13
PM19	Advance	13-May-2024	Account A	Received	O19	C15	NULL
PM20	COD	20-May-2024	Account B	Received	O20	C20	D14
PM21	COD	26-May-2024	Account B	Pending	O21	C02	D15
PM22	COD	26-May-2024	Account B	Pending	O22	C05	D16
PM23	COD	26-May-2024	Account B	Pending	O23	C08	D17
PM24	Cash	19-May-2024	Account B	Received	O24	C13	NULL
PM25	Advance	19-May-2024	Account A	Received	O25	C19	NULL
PM26	COD	01-Jun-2024	Account B	Pending	O26	C06	D18
PM27	COD	01-Jun-2024	Account B	Pending	O27	C10	D19
PM28	COD	01-Jun-2024	Account B	Pending	O28	C12	D20
PM29	Cash	25-May-2024	Account B	Received	O29	C16	NULL
PM30	COD	01-Jun-2024	Account B	Pending	O30	C18	D21

Normalization Check

1NF: Every cell has one value. PaymentID is the Primary Key. No repeating groups. PASS.

2NF: Single-column Primary Key. All attributes depend fully on PaymentID. PASS.

3NF Violation Found

A transitive dependency exists in the original Payment table:

- PaymentMethod is a non-key attribute.
- AccountUsed always depends on PaymentMethod. Every row where PaymentMethod is Advance has AccountUsed as Account A. Every row where PaymentMethod is COD or Cash has AccountUsed as Account B. This pattern is consistent across all 30 rows without a single exception.
- AccountUsed is a fact about PaymentMethod — not about the individual payment record.

Transitive dependency chain:

PaymentID -> PaymentMethod -> AccountUsed

Storing AccountUsed in Payment creates redundancy. If the account mapping ever changes, every row would need updating. This violates 3NF.

New Table Created: PaymentMethodDetail

AccountUsed is removed from Payment and placed in PaymentMethodDetail. PaymentMethod text itself is the Primary Key — no separate MethodID is needed. Payment keeps PaymentMethod as a Foreign Key.

New Table: PaymentMethodDetail (3 rows)

PaymentMethod (PK)	AccountUsed
Advance	Account A
COD	Account B
Cash	Account B

To retrieve AccountUsed when needed:

```
SELECT P.PaymentID, P.PaymentMethod, PM.AccountUsed
```

```
FROM Payment P, PaymentMethodDetail PM
```

```
WHERE P.PaymentMethod = PM.PaymentMethod;
```

3NF Violation Found — CustomerID Transitive Dependency

After removing AccountUsed in the previous normalization step, a second transitive dependency was identified in the Payment table:

- OrderID is a Foreign Key in Payment linking to the JOrder table.
- JOrder already contains CustomerID — it stores which customer placed each order.
- CustomerID in Payment is therefore not directly determined by PaymentID. It is determined through OrderID which links to JOrder which contains CustomerID.

Transitive dependency chain:

PaymentID -> OrderID -> CustomerID

Storing CustomerID in Payment is redundant because the same information is already available by joining Payment with JOrder. This is the exact same type of violation that was identified and fixed in the Refund table where RefundAccount was removed because it was reachable through another table

CustomerID Removed from Payment:

CustomerID is removed from the Payment table. When CustomerID is needed along with payment information, a JOIN between Payment and JOrder is used as shown below.

To retrieve CustomerID from Payment:

```
SELECT P. PaymentID, P. PaymentMethod, J. CustomerID
FROM Payment P, JOrder J
WHERE P.OrderID = J.OrderID;
```

Updated Payment Table After Normalization (30 rows —> AccountUsed, customerid removed)

PaymentID	PaymentMethod (FK)	PaymentDate	PaymentStatus	OrderID (FK)	DeliveryID (FK)
PM01	COD	10-May-2024	Received	O01	D01
PM02	COD	10-May-2024	Received	O02	D02
PM03	COD	10-May-2024	Received	O03	D03
PM04	COD	10-May-2024	Received	O04	D04
PM05	Advance	03-May-2024	Received	O05	NULL
PM06	COD	10-May-2024	Received	O06	D05

PM07	COD	10-May-2024	Received	O07	D06
PM08	COD	15-May-2024	Received	O08	D07
PM09	COD	15-May-2024	Received	O09	D08
PM10	Advance	08-May-2024	Received	O10	NULL
PM11	COD	15-May-2024	Received	O11	D09
PM12	COD	15-May-2024	Received	O12	D10
PM13	Advance	08-May-2024	Received	O13	NULL
PM14	COD	20-May-2024	Received	O14	D11
PM15	COD	20-May-2024	Received	O15	D12
PM16	Advance	13-May-2024	Received	O16	NULL
PM17	Advance	13-May-2024	Received	O17	NULL
PM18	COD	20-May-2024	Pending	O18	D13
PM19	Advance	13-May-2024	Received	O19	NULL
PM20	COD	20-May-2024	Received	O20	D14
PM21	COD	26-May-2024	Pending	O21	D15
PM22	COD	26-May-2024	Pending	O22	D16
PM23	COD	26-May-2024	Pending	O23	D17
PM24	Cash	19-May-2024	Received	O24	NULL
PM25	Advance	19-May-2024	Received	O25	NULL
PM26	COD	01-Jun-2024	Pending	O26	D18
PM27	COD	01-Jun-2024	Pending	O27	D19
PM28	COD	01-Jun-2024	Pending	O28	D20
PM29	Cash	25-May-2024	Received	O29	NULL
PM30	COD	01-Jun-2024	Pending	O30	D21

CONCLUSION:

Payment violated 3NF because AccountUsed was transitively dependent on PaymentMethod.

Fixed by creating PaymentMethodDetail. AccountUsed removed from Payment.

Payment table had a second 3NF violation. CustomerID was transitively dependent on PaymentID through OrderID.

Since JOrder already stores CustomerID, keeping it in Payment created redundancy. CustomerID removed from Payment. Retrieved via JOIN with JOrder when needed.

Table 7: Delivery (Weak Entity)

DeliveryID (PK)	TrackingNumber	DeliveryStatus	DeliveryDate	OrderID (FK)
D01	TCS-2024-05001	Delivered	08-May-2024	O01
D02	TCS-2024-05002	Delivered	08-May-2024	O02
D03	TCS-2024-05003	Delivered	09-May-2024	O03
D04	TCS-2024-05004	Delivered	09-May-2024	O04
D05	TCS-2024-05005	Delivered	09-May-2024	O06
D06	TCS-2024-05006	Delivered	10-May-2024	O07
D07	TCS-2024-05007	Delivered	14-May-2024	O08
D08	TCS-2024-05008	Delivered	14-May-2024	O09
D09	TCS-2024-05009	Delivered	14-May-2024	O11
D10	TCS-2024-05010	Delivered	15-May-2024	O12
D11	TCS-2024-05011	Delivered	19-May-2024	O14
D12	TCS-2024-05012	Delivered	19-May-2024	O15
D13	TCS-2024-05013	Pending	NULL	O18
D14	TCS-2024-05014	Delivered	20-May-2024	O20
D15	TCS-2024-05015	Pending	NULL	O21
D16	TCS-2024-05016	Pending	NULL	O22
D17	TCS-2024-05017	Pending	NULL	O23
D18	TCS-2024-05018	Pending	NULL	O26
D19	TCS-2024-05019	Pending	NULL	O27
D20	TCS-2024-05020	Pending	NULL	O28

D21	TCS-2024-05021	Pending	NULL	O30
-----	----------------	---------	------	-----

Interpretation:

The Delivery entity stores shipment information such as tracking number, delivery status, delivery date, and related order ID. DeliveryID uniquely identifies each delivery record and all attributes depend directly on it. There are no repeating groups, partial dependencies, or transitive dependencies in the table.

Conclusion:

Delivery is already in 3NF. No normalization required.

Table 8: Defect (Weak Entity)

DefectID (PK)	DefectDescription	DefectDate	OrderID (FK)	ProductID (FK)
DF01	Wrong size delivered - ordered size 8, received size 10	11-May-2024	O04	P09
DF02	Fabric torn at seam on Khaadi Embroidered 3PC Khaddar	12-May-2024	O06	P06
DF03	Missing piece - 3PC suit had only 2 pieces in parcel	21-May-2024	O15	P03
DF04	Color fading - product not matching live session display	22-May-2024	O14	P08
DF05	Wrong item delivered - received Generation instead of Khaadi	23-May-2024	O11	P06

Interpretation:

The Defect entity records customer complaints or product issues related to orders. Each defect record contains a unique DefectID along with defect description, defect date, order ID, and product ID. All attributes are directly dependent on the primary key without depending on each other.

CONCLUSION:

Defect is already in 3NF. No normalization required.

Table 9: Refund (Weak Entity) —> Normalization Required

Original Refund Table Before Normalization (5 rows matching all 5 defects)

RefundID (PK)	RefundDate	RefundAmount	RefundAccount	RefundStatus	DefectID (FK)	PaymentID (FK)
R01	14-May-2024	6200	Account B	Processed	DF01	PM04
R02	15-May-2024	7800	Account B	Processed	DF02	PM06
R03	24-May-2024	9600	Account B	Pending	DF03	PM15
R04	26-May-2024	4800	Account B	Pending	DF04	PM14
R05	28-May-2024	7800	Account B	Pending	DF05	PM11

Normalization Check

1NF: Every cell has one value. RefundID is the Primary Key. No repeating groups. PASS.

2NF: Single-column Primary Key. All attributes depend fully on RefundID. PASS.

3NF Violation Found

A transitive dependency exists in the original Refund table:

- PaymentID is a Foreign Key linking Refund to Payment.
- RefundAccount stores which bank account to refund from — Account A or Account B.
- RefundAccount is always determined by the PaymentMethod of the linked payment. Advance payments refund from Account A. COD and Cash payments refund from Account B.
- All 5 refund rows have Account B because all 5 linked payments were COD. This proves RefundAccount is not independently determined by RefundID — it follows the payment method.

Transitive dependency chain:

RefundID -> PaymentID -> PaymentMethod -> AccountUsed = RefundAccount

RefundAccount duplicates information already available in PaymentMethodDetail created during Payment normalization. This violates 3NF.

RefundAccount Removed

RefundAccount is removed from Refund entirely. The account information is retrieved by joining Refund with Payment and PaymentMethodDetail when needed.

Updated Refund Table After Normalization (5 rows → RefundAccount removed)

RefundID (PK)	RefundDate	RefundAmount	RefundStatus	DefectID (FK)	PaymentID (FK)
R01	14-May-2024	6200	Processed	DF01	PM04
R02	15-May-2024	7800	Processed	DF02	PM06
R03	24-May-2024	9600	Pending	DF03	PM15
R04	26-May-2024	4800	Pending	DF04	PM14
R05	28-May-2024	7800	Pending	DF05	PM11

To retrieve refund account when needed:

SELECT R.RefundID, R.RefundAmount, PM.AccountUsed

FROM Refund R, Payment P, PaymentMethodDetail PM

WHERE R.PaymentID = P.PaymentID

AND P.PaymentMethod = PM.PaymentMethod;

CONCLUSION:

Refund violated 3NF because RefundAccount was transitively dependent on PaymentMethod through PaymentID. Fixed by removing RefundAccount. Retrieved via JOIN when needed.

Final Normalization Summary:

Table	1NF	2NF	3NF	Action Taken
Supplier	Pass	Pass	Pass	No normalization required
Customer	Pass	Pass	Pass	No normalization required
Product	Pass	Pass	Fail	ProductStitched and ProductUnstitched tables created. Each stores only ProductID and its respective attribute (StitchedSize or UnstitchedPieces). ProductName not repeated — already in main Product table. ProductQuantity remains in Product.
Order	Pass	Pass	Pass	No normalization required
OrderProduct	Pass	Pass	Pass	No normalization required. Quantity renamed to OrderQuantity.
Payment	Pass	Pass	Fail	PaymentMethodDetail table created. AccountUsed removed from Payment.
Delivery	Pass	Pass	Pass	No normalization required
Defect	Pass	Pass	Pass	No normalization required
Refund	Pass	Pass	Fail	RefundAccount removed from Refund. Retrieved via JOIN with PaymentMethodDetail.

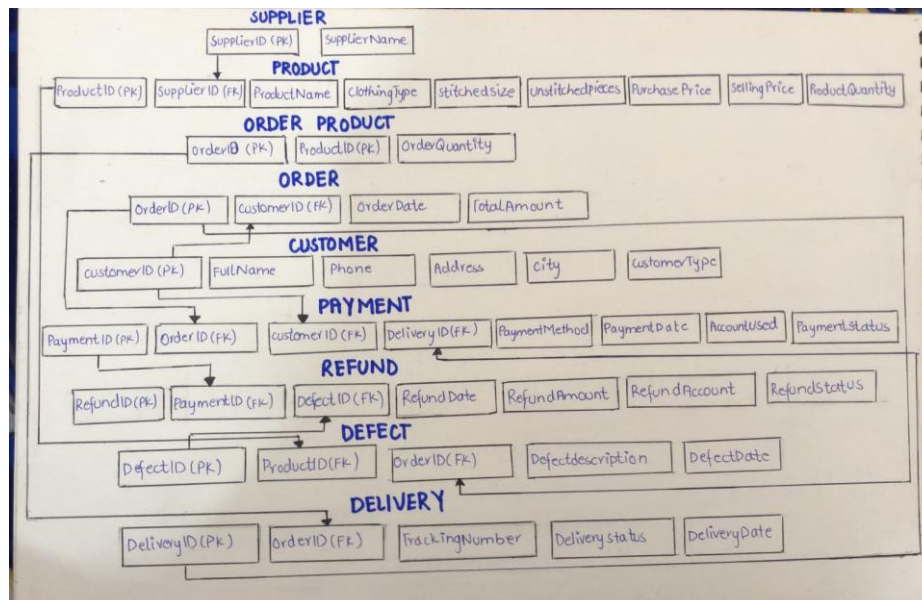
1. Tables that required normalization: **3 (Product, Payment, Refund)**
2. Tables already in 3NF: **6 (Supplier, Customer, Order, OrderProduct, Delivery, Defect)**
3. New tables created: **3 (ProductStitched, ProductUnstitched, PaymentMethodDetail)**
4. Total tables in final normalized database: **12**

4. RELATIONAL SCHEMA SUMMARY:

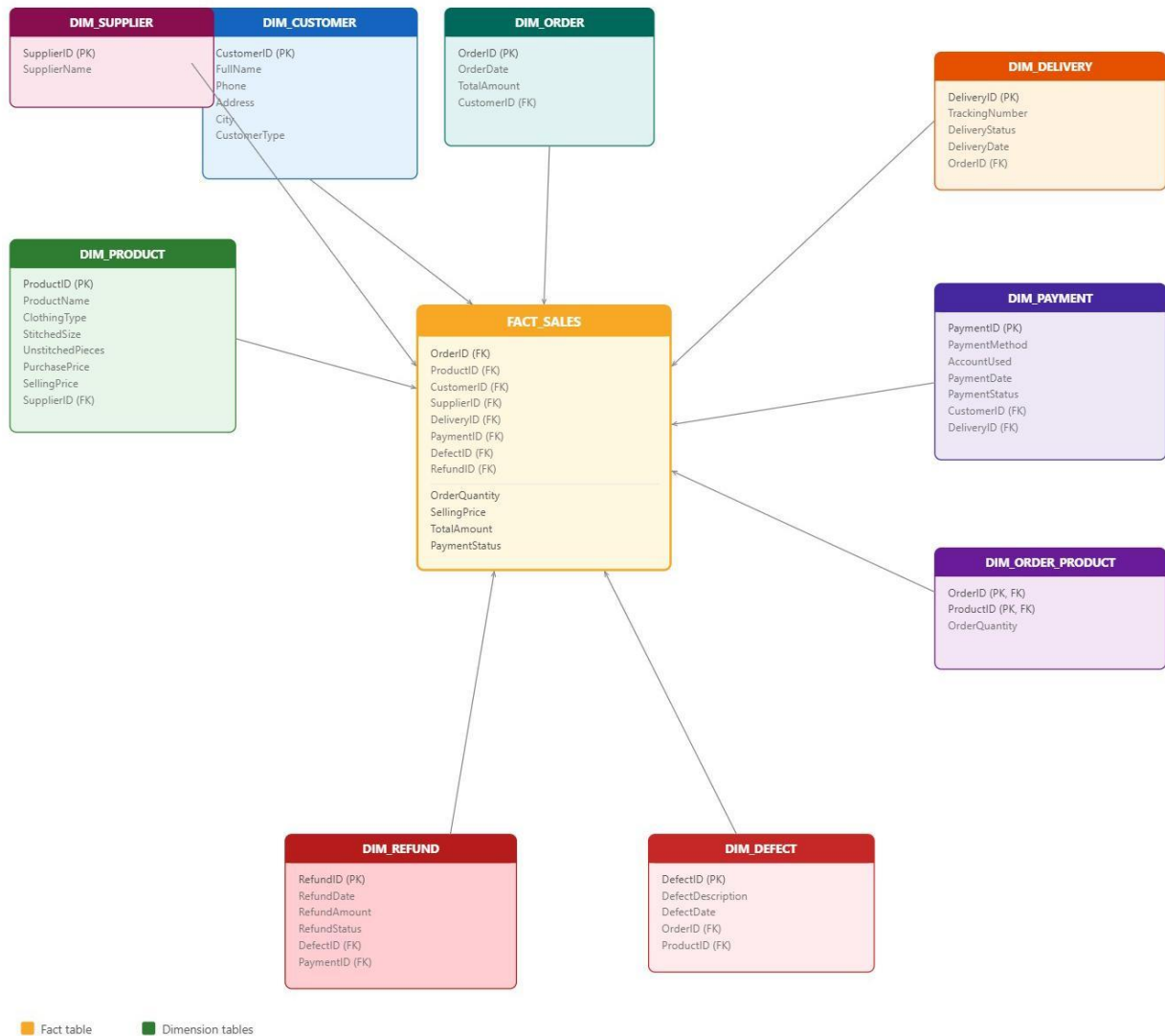
This formal schema shows our full, normalized database structure:

- **Supplier** (**SupplierID**, SupplierName)
- **Customer** (**CustomerID**, FullName, Phone, Address, City, CustomerType)
- **Product** (**ProductID**, ProductName, ClothingType, PurchasePrice, SellingPrice, ProductQuantity, PurchaseDate, *SupplierID*)
- **ProductStitched** (*ProductID*, StitchedSize)
- **ProductUnstitched** (*ProductID*, UnstitchedPieces)
- **Order** (**OrderID**, OrderDate, TotalAmount, *CustomerID*)
- **OrderProduct** (**OrderID**, **ProductID**, OrderQuantity)
- **PaymentMethodDetail** (**PaymentMethod**, AccountUsed)
- **Payment** (**PaymentID**, *PaymentMethod*, PaymentDate, PaymentStatus, *OrderID*, *CustomerID*, *DeliveryID*)
- **Delivery** (**DeliveryID**, TrackingNumber, DeliveryStatus, DeliveryDate, *OrderID*)
- **Defect** (**DefectID**, DefectDescription, DefectDate, *OrderID*, *ProductID*)
- **Refund** (**RefundID**, RefundDate, RefundAmount, RefundStatus, *DefectID*, *PaymentID*)

Bold text indicates Primary Keys; *Italics* indicate Foreign Keys.



Star Schema:

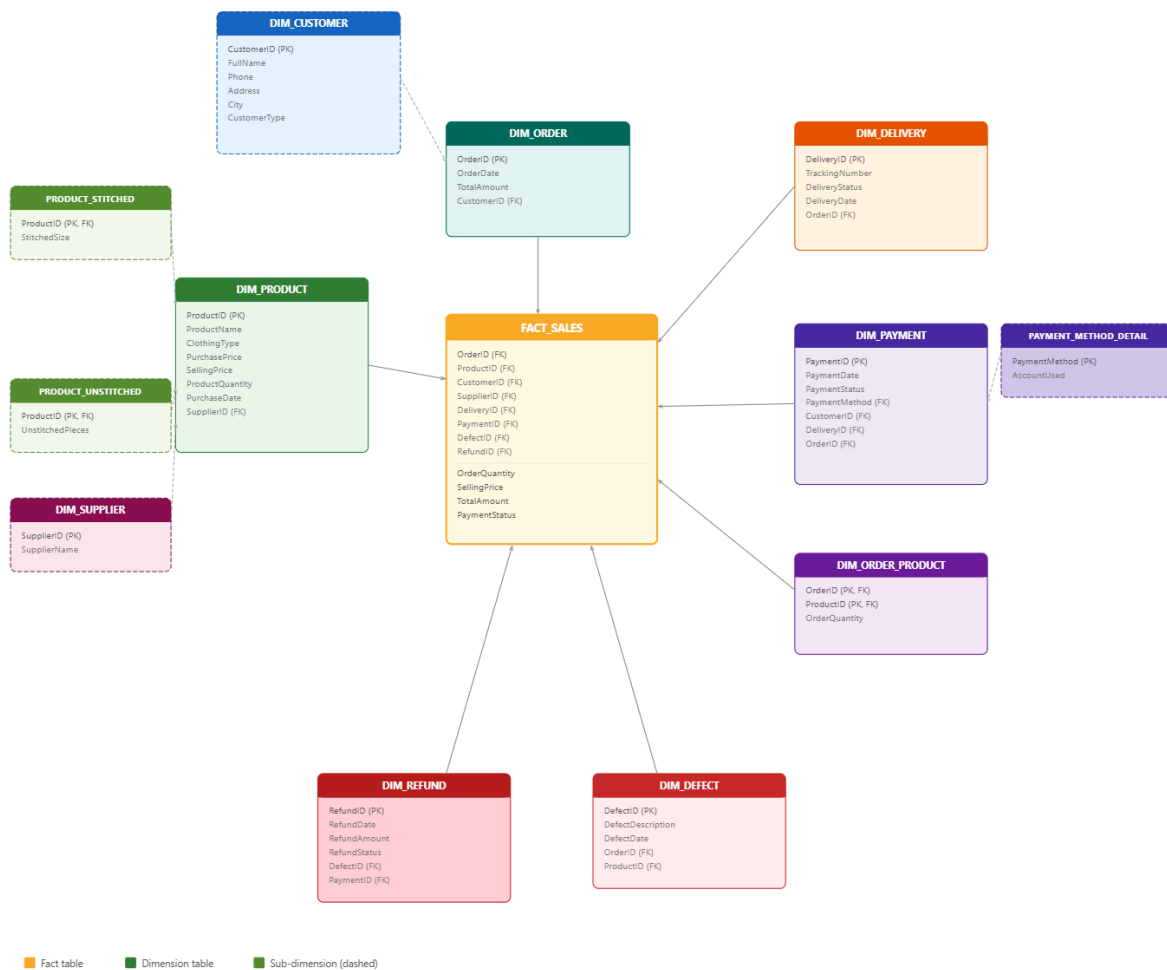


- **Centralized Fact Focus:** The design centers around a main **Fact Sales** table that stores measurable, numerical metrics such as quantities sold, total revenue, and calculated profit margins generated during Facebook Live sessions.
- **Denormalized Dimensions:** The fact table is directly surrounded by a single layer of flat, denormalized dimension tables like DimCustomer, DimProduct, and DimTime, which contain descriptive data without further sub-tables.

- **Query Performance Optimization:** By eliminating complex table joins, this star-like structure is built specifically for speed, allowing you to quickly run business analytics and generate fast sales dashboards.

Snowflake Schema:

Snowflake schema — J. Alam Business Database



- **Normalized Dimension Hierarchies:** Unlike the star schema, this model splits low-level attributes into further sub-tables, breaking down dimensions into a fully normalized, branching structure.

- **Structural Data Integrity:** By isolating multi-level details—such as branching DimProduct into a separate DimSupplier table, or DimCustomer into a separate DimCity table—it completely eliminates redundant text storage.
- **Storage Optimization:** This design keeps the database highly structured and minimizes storage waste, though it requires more complex multi-table joins when running analytical queries.

6. Codes:

CODE 1: SUPPLIER

```
CREATE TABLE Supplier (
    SupplierID VARCHAR2(5),
    SupplierName VARCHAR2(30) NOT NULL,
    CONSTRAINT Supplier_SupplierID_PK PRIMARY KEY (SupplierID)
);

INSERT INTO Supplier VALUES ('S01', 'Sapphire');
INSERT INTO Supplier VALUES ('S02', 'Khaadi');
INSERT INTO Supplier VALUES ('S03', 'Generation');
INSERT INTO Supplier VALUES ('S04', 'Ethnic by Outfitters');
```

CODE 2: CUSTOMER

```
CREATE TABLE Customer (
    CustomerID VARCHAR2(5),
    FullName VARCHAR2(30) NOT NULL,
    Phone VARCHAR2(15) NOT NULL,
    Address VARCHAR2(100) NOT NULL,
    City VARCHAR2(20) NOT NULL,
    CustomerType VARCHAR2(5) NOT NULL,
    CONSTRAINT Customer_CustomerID_PK PRIMARY KEY (CustomerID),
    CONSTRAINT Customer_Phone_UQ UNIQUE (Phone),
    CONSTRAINT Customer_CustomerType_CHK CHECK (CustomerType IN ('Old', 'New'))
);

INSERT INTO Customer VALUES ('C01', 'Ayesha Tariq', '03001234567', 'House 12 Block B Gulberg III', 'Lahore', 'Old');
INSERT INTO Customer VALUES ('C02', 'Fatima Malik', '03111234568', 'Flat 5 Clifton Block 4', 'Karachi', 'Old');
INSERT INTO Customer VALUES ('C03', 'Sara Hussain', '03211234569', 'House 7 F-8 Markaz', 'Islamabad', 'Old');
INSERT INTO Customer VALUES ('C04', 'Zara Ahmed', '03311234560', 'House 3 Hayatabad Phase 2', 'Peshawar', 'New');
INSERT INTO Customer VALUES ('C05', 'Hina Baig', '03421234561', 'Flat 9 Bahria Town Phase 4', 'Rawalpindi', 'Old');
INSERT INTO Customer VALUES ('C06', 'Sana Qureshi', '03501234562', 'House 22 DHA Phase 6', 'Lahore', 'Old');
INSERT INTO Customer VALUES ('C07', 'Maria Iqbal', '03021234563', 'Flat 2 North Nazimabad Block H', 'Karachi', 'New');
INSERT INTO Customer VALUES ('C08', 'Nadia Raza', '03121234564', 'House 45 G-10 Markaz', 'Islamabad', 'Old');
INSERT INTO Customer VALUES ('C09', 'Asma Khan', '03221234565', 'House 8 Model Town', 'Lahore', 'New');
INSERT INTO Customer VALUES ('C10', 'Rabia Ali', '03321234566', 'Flat 11 Defence Phase 2', 'Karachi', 'Old');
INSERT INTO Customer VALUES ('C11', 'Ume Kulsoom', '03401234570', 'House 6 Johar Town', 'Lahore', 'New');
INSERT INTO Customer VALUES ('C12', 'Bushra Noor', '03451234571', 'Flat 14 Gulshan-e-Iqbal Block 3', 'Karachi', 'Old');
INSERT INTO Customer VALUES ('C13', 'Amna Siddiqui', '03061234572', 'House 18 E-7 Islamabad', 'Islamabad', 'New');
INSERT INTO Customer VALUES ('C14', 'Rida Farooq', '03161234573', 'House 9 Saddar Cantonment', 'Rawalpindi', 'Old');
INSERT INTO Customer VALUES ('C15', 'Maham Riaz', '03261234574', 'Flat 3 Bahria Town Precinct 12', 'Karachi', 'New');
INSERT INTO Customer VALUES ('C16', 'Iqra Shahid', '03361234575', 'House 33 Cavalry Ground', 'Lahore', 'Old');
INSERT INTO Customer VALUES ('C17', 'Nimra Javed', '03461234576', 'House 5 Satellite Town', 'Rawalpindi', 'New');
INSERT INTO Customer VALUES ('C18', 'Kiran Nasir', '03071234577', 'Flat 7 F-11 Markaz', 'Islamabad', 'Old');
INSERT INTO Customer VALUES ('C19', 'Sadia Pervaiz', '03171234578', 'House 2 Wapda Town Phase 1', 'Lahore', 'New');
INSERT INTO Customer VALUES ('C20', 'Maryam Zahid', '03271234579', 'Flat 6 Block 13 Gulshan', 'Karachi', 'Old');
```


CODE 3: PRODUCT

```

CREATE TABLE Product (
  ProductID VARCHAR2(5),
  ProductName VARCHAR2(40) NOT NULL,
  ClothingType VARCHAR2(10) NOT NULL,
  PurchasePrice NUMBER(10,2) NOT NULL,
  SellingPrice NUMBER(10,2) NOT NULL,
  ProductQuantity NUMBER(4) NOT NULL,
  PurchaseDate DATE NOT NULL,
  SupplierID VARCHAR2(5) NOT NULL,
  CONSTRAINT Product_ProductID_PK PRIMARY KEY (ProductID),
  CONSTRAINT Product_SupplierID_FK FOREIGN KEY (SupplierID) REFERENCES Supplier(SupplierID),
  CONSTRAINT Product_ClothingType_CHK CHECK (ClothingType IN ('Stitched', 'Unstitched')),
  CONSTRAINT Product_PurchasePrice_CHK CHECK (PurchasePrice > 0),
  CONSTRAINT Product_SellingPrice_CHK CHECK (SellingPrice > 0),
  CONSTRAINT Product_Quantity_CHK CHECK (ProductQuantity >= 0)
);

INSERT INTO Product VALUES ('P01', 'Sapphire Embroidered 3PC Lawn Suit', 'Unstitched', 5500, 7000, 8, TO_DATE('12-02-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P02', 'Sapphire Printed 2PC Lawn', 'Unstitched', 3800, 4800, 12, TO_DATE('12-02-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P03', 'Sapphire Lawn Stitched Shirt', 'Stitched', 2800, 3500, 15, TO_DATE('15-02-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P04', 'Sapphire Khaddar Stitched Kurta', 'Stitched', 2400, 3000, 10, TO_DATE('15-02-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P05', 'Sapphire Winter Embroidered 1PC', 'Unstitched', 2200, 2800, 20, TO_DATE('20-02-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P06', 'Khaadi Embroidered 3PC Khaddar', 'Unstitched', 6000, 7800, 6, TO_DATE('03-03-2024','DD-MM-YYYY'), 'S02');
INSERT INTO Product VALUES ('P07', 'Khaadi Printed 2PC Lawn', 'Unstitched', 4000, 5200, 14, TO_DATE('03-03-2024','DD-MM-YYYY'), 'S02');
INSERT INTO Product VALUES ('P08', 'Khaadi Khaddar Stitched Kurta', 'Stitched', 3200, 4000, 9, TO_DATE('05-03-2024','DD-MM-YYYY'), 'S02');
INSERT INTO Product VALUES ('P09', 'Khaadi Stitched Shirt', 'Stitched', 2600, 3300, 11, TO_DATE('05-03-2024','DD-MM-YYYY'), 'S02');
INSERT INTO Product VALUES ('P10', 'Khaadi Printed 1PC Dupatta', 'Unstitched', 1500, 2000, 25, TO_DATE('10-03-2024','DD-MM-YYYY'), 'S02');
INSERT INTO Product VALUES ('P11', 'Generation Embroidered 3PC', 'Unstitched', 4800, 6200, 7, TO_DATE('18-03-2024','DD-MM-YYYY'), 'S03');
INSERT INTO Product VALUES ('P12', 'Generation Printed 1PC Dupatta', 'Unstitched', 1800, 2400, 18, TO_DATE('18-03-2024','DD-MM-YYYY'), 'S03');
INSERT INTO Product VALUES ('P13', 'Generation Stitched Linen Shirt', 'Stitched', 2500, 3200, 13, TO_DATE('20-03-2024','DD-MM-YYYY'), 'S03');
INSERT INTO Product VALUES ('P14', 'Generation Lawn 2PC Suit', 'Unstitched', 3500, 4500, 9, TO_DATE('22-03-2024','DD-MM-YYYY'), 'S03');

INSERT INTO Product VALUES ('P15', 'Generation Stitched Formal Kurta', 'Stitched', 3000, 3800, 8, TO_DATE('25-03-2024','DD-MM-YYYY'), 'S03');
INSERT INTO Product VALUES ('P16', 'Ethnic Printed Lawn 3PC', 'Unstitched', 3800, 4800, 10, TO_DATE('10-04-2024','DD-MM-YYYY'), 'S04');
INSERT INTO Product VALUES ('P17', 'Ethnic Embroidered 2PC', 'Unstitched', 3500, 4500, 12, TO_DATE('10-04-2024','DD-MM-YYYY'), 'S04');
INSERT INTO Product VALUES ('P18', 'Ethnic Stitched Casual Kurta', 'Stitched', 2000, 2600, 16, TO_DATE('12-04-2024','DD-MM-YYYY'), 'S04');
INSERT INTO Product VALUES ('P19', 'Ethnic Printed 1PC Shirt', 'Unstitched', 1600, 2100, 22, TO_DATE('12-04-2024','DD-MM-YYYY'), 'S04');
INSERT INTO Product VALUES ('P20', 'Ethnic Stitched Khaddar Kurta', 'Stitched', 2800, 3500, 7, TO_DATE('15-04-2024','DD-MM-YYYY'), 'S04');
INSERT INTO Product VALUES ('P21', 'Sapphire Summer Printed 2PC', 'Unstitched', 4200, 5400, 11, TO_DATE('18-04-2024','DD-MM-YYYY'), 'S01');
INSERT INTO Product VALUES ('P22', 'Khaadi Summer Stitched Shirt', 'Stitched', 2700, 3400, 9, TO_DATE('20-04-2024','DD-MM-YYYY'), 'S02');

```

CODE 4: PRODUCT STITCHED

```

CREATE TABLE ProductStitched (
  ProductID VARCHAR2(5),
  StitchedSize VARCHAR2(5) NOT NULL,
  CONSTRAINT ProductStitched_ProductID_PK PRIMARY KEY (ProductID),
  CONSTRAINT ProductStitched_ProductID_FK FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
  CONSTRAINT ProductStitched_Size_CHK CHECK (StitchedSize IN ('8', '10', '12', '14', '16'))
);

INSERT INTO ProductStitched VALUES ('P03', '10');
INSERT INTO ProductStitched VALUES ('P04', '12');
INSERT INTO ProductStitched VALUES ('P08', '14');
INSERT INTO ProductStitched VALUES ('P09', '8');
INSERT INTO ProductStitched VALUES ('P13', '16');
INSERT INTO ProductStitched VALUES ('P15', '10');
INSERT INTO ProductStitched VALUES ('P18', '10');
INSERT INTO ProductStitched VALUES ('P20', '12');
INSERT INTO ProductStitched VALUES ('P22', '14');

```

CODE 5: PRODUCT UNSTITCHED

```

CREATE TABLE ProductUnstitched (
    ProductID VARCHAR2(5),
    UnstitchedPieces VARCHAR2(10) NOT NULL,
    CONSTRAINT ProductUnstitched_ProductID_PK PRIMARY KEY (ProductID),
    CONSTRAINT ProductUnstitched_ProductID_FK FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
    CONSTRAINT ProductUnstitched_Pieces_CHK CHECK (UnstitchedPieces IN ('1 Piece', '2 Piece', '3 Piece'))
);

INSERT INTO ProductUnstitched VALUES ('P01', '3 Piece');
INSERT INTO ProductUnstitched VALUES ('P02', '2 Piece');
INSERT INTO ProductUnstitched VALUES ('P05', '1 Piece');
INSERT INTO ProductUnstitched VALUES ('P06', '3 Piece');
INSERT INTO ProductUnstitched VALUES ('P07', '2 Piece');
INSERT INTO ProductUnstitched VALUES ('P10', '1 Piece');
INSERT INTO ProductUnstitched VALUES ('P11', '3 Piece');
INSERT INTO ProductUnstitched VALUES ('P12', '1 Piece');
INSERT INTO ProductUnstitched VALUES ('P14', '2 Piece');
INSERT INTO ProductUnstitched VALUES ('P16', '3 Piece');
INSERT INTO ProductUnstitched VALUES ('P17', '2 Piece');
INSERT INTO ProductUnstitched VALUES ('P19', '1 Piece');
INSERT INTO ProductUnstitched VALUES ('P21', '2 Piece');

```

CODE 6: JORDER

```

CREATE TABLE JOrder (
    OrderID VARCHAR2(5),
    OrderDate DATE NOT NULL,
    TotalAmount NUMBER(10,2) NOT NULL,
    CustomerID VARCHAR2(5) NOT NULL,
    CONSTRAINT JOrder_OrderID_PK PRIMARY KEY (OrderID),
    CONSTRAINT JOrder_CustomerID_FK FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    CONSTRAINT JOrder_TotalAmount_CHK CHECK (TotalAmount > 0)
);

INSERT INTO JOrder VALUES ('001', TO_DATE('03-05-2024','DD-MM-YYYY'), 3500, 'C01');
INSERT INTO JOrder VALUES ('002', TO_DATE('03-05-2024','DD-MM-YYYY'), 3000, 'C03');
INSERT INTO JOrder VALUES ('003', TO_DATE('03-05-2024','DD-MM-YYYY'), 9200, 'C05');
INSERT INTO JOrder VALUES ('004', TO_DATE('03-05-2024','DD-MM-YYYY'), 6500, 'C08');
INSERT INTO JOrder VALUES ('005', TO_DATE('03-05-2024','DD-MM-YYYY'), 3500, 'C12');
INSERT INTO JOrder VALUES ('006', TO_DATE('03-05-2024','DD-MM-YYYY'), 7800, 'C16');
INSERT INTO JOrder VALUES ('007', TO_DATE('03-05-2024','DD-MM-YYYY'), 5600, 'C18');
INSERT INTO JOrder VALUES ('008', TO_DATE('08-05-2024','DD-MM-YYYY'), 3000, 'C02');
INSERT INTO JOrder VALUES ('009', TO_DATE('08-05-2024','DD-MM-YYYY'), 3500, 'C06');
INSERT INTO JOrder VALUES ('010', TO_DATE('08-05-2024','DD-MM-YYYY'), 7300, 'C09');
INSERT INTO JOrder VALUES ('011', TO_DATE('08-05-2024','DD-MM-YYYY'), 13000, 'C10');
INSERT INTO JOrder VALUES ('012', TO_DATE('08-05-2024','DD-MM-YYYY'), 3200, 'C14');
INSERT INTO JOrder VALUES ('013', TO_DATE('08-05-2024','DD-MM-YYYY'), 2600, 'C17');
INSERT INTO JOrder VALUES ('014', TO_DATE('13-05-2024','DD-MM-YYYY'), 8000, 'C01');
INSERT INTO JOrder VALUES ('015', TO_DATE('13-05-2024','DD-MM-YYYY'), 13500, 'C03');
INSERT INTO JOrder VALUES ('016', TO_DATE('13-05-2024','DD-MM-YYYY'), 5200, 'C04');
INSERT INTO JOrder VALUES ('017', TO_DATE('13-05-2024','DD-MM-YYYY'), 3300, 'C07');
INSERT INTO JOrder VALUES ('018', TO_DATE('13-05-2024','DD-MM-YYYY'), 4800, 'C11');
INSERT INTO JOrder VALUES ('019', TO_DATE('13-05-2024','DD-MM-YYYY'), 3200, 'C15');
INSERT INTO JOrder VALUES ('020', TO_DATE('13-05-2024','DD-MM-YYYY'), 6200, 'C20');
INSERT INTO JOrder VALUES ('021', TO_DATE('19-05-2024','DD-MM-YYYY'), 7800, 'C02');
INSERT INTO JOrder VALUES ('022', TO_DATE('19-05-2024','DD-MM-YYYY'), 3500, 'C05');

INSERT INTO JOrder VALUES ('023', TO_DATE('19-05-2024','DD-MM-YYYY'), 8000, 'C08');
INSERT INTO JOrder VALUES ('024', TO_DATE('19-05-2024','DD-MM-YYYY'), 2400, 'C13');
INSERT INTO JOrder VALUES ('025', TO_DATE('19-05-2024','DD-MM-YYYY'), 4500, 'C19');
INSERT INTO JOrder VALUES ('026', TO_DATE('25-05-2024','DD-MM-YYYY'), 6200, 'C06');
INSERT INTO JOrder VALUES ('027', TO_DATE('25-05-2024','DD-MM-YYYY'), 5200, 'C10');
INSERT INTO JOrder VALUES ('028', TO_DATE('25-05-2024','DD-MM-YYYY'), 7800, 'C12');
INSERT INTO JOrder VALUES ('029', TO_DATE('25-05-2024','DD-MM-YYYY'), 3500, 'C16');
INSERT INTO JOrder VALUES ('030', TO_DATE('25-05-2024','DD-MM-YYYY'), 3200, 'C18');

```


CODE 7: ORDERPRODUCT

```

CREATE TABLE OrderProduct (
  OrderID      VARCHAR2(5),
  ProductID    VARCHAR2(5),
  OrderQuantity NUMBER(4) NOT NULL,
  CONSTRAINT OrderProduct_PK PRIMARY KEY (OrderID, ProductID),
  CONSTRAINT OrderProduct_OrderID_FK FOREIGN KEY (OrderID) REFERENCES JOrder(OrderID),
  CONSTRAINT OrderProduct_ProductID_FK FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
  CONSTRAINT OrderProduct_Quantity_CHK CHECK (OrderQuantity > 0)
);

INSERT INTO OrderProduct VALUES ('001', 'P03', 1);
INSERT INTO OrderProduct VALUES ('002', 'P04', 1);
INSERT INTO OrderProduct VALUES ('003', 'P08', 1);
INSERT INTO OrderProduct VALUES ('003', 'P07', 1);
INSERT INTO OrderProduct VALUES ('004', 'P09', 1);
INSERT INTO OrderProduct VALUES ('004', 'P13', 1);
INSERT INTO OrderProduct VALUES ('005', 'P03', 1);
INSERT INTO OrderProduct VALUES ('006', 'P06', 1);
INSERT INTO OrderProduct VALUES ('007', 'P18', 1);
INSERT INTO OrderProduct VALUES ('007', 'P04', 1);
INSERT INTO OrderProduct VALUES ('008', 'P04', 1);
INSERT INTO OrderProduct VALUES ('009', 'P03', 1);
INSERT INTO OrderProduct VALUES ('010', 'P08', 1);
INSERT INTO OrderProduct VALUES ('010', 'P09', 1);
INSERT INTO OrderProduct VALUES ('011', 'P06', 1);
INSERT INTO OrderProduct VALUES ('011', 'P07', 1);
INSERT INTO OrderProduct VALUES ('012', 'P13', 1);
INSERT INTO OrderProduct VALUES ('013', 'P18', 1);
INSERT INTO OrderProduct VALUES ('014', 'P08', 2);
INSERT INTO OrderProduct VALUES ('015', 'P03', 3);
INSERT INTO OrderProduct VALUES ('015', 'P04', 1);

INSERT INTO OrderProduct VALUES ('016', 'P07', 1);
INSERT INTO OrderProduct VALUES ('017', 'P09', 1);
INSERT INTO OrderProduct VALUES ('018', 'P16', 1);
INSERT INTO OrderProduct VALUES ('019', 'P13', 1);
INSERT INTO OrderProduct VALUES ('020', 'P11', 1);
INSERT INTO OrderProduct VALUES ('021', 'P06', 1);
INSERT INTO OrderProduct VALUES ('022', 'P03', 1);
INSERT INTO OrderProduct VALUES ('023', 'P08', 2);
INSERT INTO OrderProduct VALUES ('024', 'P12', 1);
INSERT INTO OrderProduct VALUES ('025', 'P17', 1);
INSERT INTO OrderProduct VALUES ('026', 'P11', 1);
INSERT INTO OrderProduct VALUES ('027', 'P18', 2);
INSERT INTO OrderProduct VALUES ('028', 'P06', 1);
INSERT INTO OrderProduct VALUES ('029', 'P03', 1);
INSERT INTO OrderProduct VALUES ('030', 'P13', 1);

```

CODE 8: PAYMENT

```

CREATE TABLE Payment (
  PaymentID      VARCHAR2(6),
  PaymentMethod  VARCHAR2(10) NOT NULL,
  PaymentDate    DATE NOT NULL,
  PaymentStatus  VARCHAR2(10) NOT NULL,
  OrderID        VARCHAR2(5) NOT NULL,
  DeliveryID     VARCHAR2(5),
  CONSTRAINT Payment_PaymentID_PK PRIMARY KEY (PaymentID),
  CONSTRAINT Payment_OrderID_UQ UNIQUE (OrderID),
  CONSTRAINT Payment_PaymentMethod_FK FOREIGN KEY (PaymentMethod) REFERENCES PaymentMethodDetail(PaymentMethod),
  CONSTRAINT Payment_OrderID_FK FOREIGN KEY (OrderID) REFERENCES JOrder(OrderID),
  CONSTRAINT Payment_PaymentStatus_CHK CHECK (PaymentStatus IN ('Received', 'Pending'))
);

```

```

INSERT INTO Payment VALUES ('PM01', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '001', 'D01');
INSERT INTO Payment VALUES ('PM02', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '002', 'D02');
INSERT INTO Payment VALUES ('PM03', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '003', 'D03');
INSERT INTO Payment VALUES ('PM04', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '004', 'D04');
INSERT INTO Payment VALUES ('PM05', 'Advance', TO_DATE('03-05-2024', 'DD-MM-YYYY'), 'Received', '005', NULL);
INSERT INTO Payment VALUES ('PM06', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '006', 'D05');
INSERT INTO Payment VALUES ('PM07', 'COD', TO_DATE('10-05-2024', 'DD-MM-YYYY'), 'Received', '007', 'D06');
INSERT INTO Payment VALUES ('PM08', 'COD', TO_DATE('15-05-2024', 'DD-MM-YYYY'), 'Received', '008', 'D07');
INSERT INTO Payment VALUES ('PM09', 'COD', TO_DATE('15-05-2024', 'DD-MM-YYYY'), 'Received', '009', 'D08');
INSERT INTO Payment VALUES ('PM10', 'Advance', TO_DATE('08-05-2024', 'DD-MM-YYYY'), 'Received', '010', NULL);
INSERT INTO Payment VALUES ('PM11', 'COD', TO_DATE('15-05-2024', 'DD-MM-YYYY'), 'Received', '011', 'D09');
INSERT INTO Payment VALUES ('PM12', 'COD', TO_DATE('15-05-2024', 'DD-MM-YYYY'), 'Received', '012', 'D10');
INSERT INTO Payment VALUES ('PM13', 'Advance', TO_DATE('08-05-2024', 'DD-MM-YYYY'), 'Received', '013', NULL);
INSERT INTO Payment VALUES ('PM14', 'COD', TO_DATE('20-05-2024', 'DD-MM-YYYY'), 'Received', '014', 'D11');
INSERT INTO Payment VALUES ('PM15', 'COD', TO_DATE('20-05-2024', 'DD-MM-YYYY'), 'Received', '015', 'D12');
INSERT INTO Payment VALUES ('PM16', 'Advance', TO_DATE('13-05-2024', 'DD-MM-YYYY'), 'Received', '016', NULL);
INSERT INTO Payment VALUES ('PM17', 'Advance', TO_DATE('13-05-2024', 'DD-MM-YYYY'), 'Received', '017', NULL);
INSERT INTO Payment VALUES ('PM18', 'COD', TO_DATE('20-05-2024', 'DD-MM-YYYY'), 'Pending', '018', 'D13');
INSERT INTO Payment VALUES ('PM19', 'Advance', TO_DATE('13-05-2024', 'DD-MM-YYYY'), 'Received', '019', NULL);
INSERT INTO Payment VALUES ('PM20', 'COD', TO_DATE('20-05-2024', 'DD-MM-YYYY'), 'Received', '020', 'D14');
INSERT INTO Payment VALUES ('PM21', 'COD', TO_DATE('26-05-2024', 'DD-MM-YYYY'), 'Pending', '021', 'D15');
INSERT INTO Payment VALUES ('PM22', 'COD', TO_DATE('26-05-2024', 'DD-MM-YYYY'), 'Pending', '022', 'D16');
INSERT INTO Payment VALUES ('PM23', 'COD', TO_DATE('26-05-2024', 'DD-MM-YYYY'), 'Pending', '023', 'D17');
INSERT INTO Payment VALUES ('PM24', 'Cash', TO_DATE('19-05-2024', 'DD-MM-YYYY'), 'Received', '024', NULL);
INSERT INTO Payment VALUES ('PM25', 'Advance', TO_DATE('19-05-2024', 'DD-MM-YYYY'), 'Received', '025', NULL);
INSERT INTO Payment VALUES ('PM26', 'COD', TO_DATE('01-06-2024', 'DD-MM-YYYY'), 'Pending', '026', 'D18');
INSERT INTO Payment VALUES ('PM27', 'COD', TO_DATE('01-06-2024', 'DD-MM-YYYY'), 'Pending', '027', 'D19');
INSERT INTO Payment VALUES ('PM28', 'COD', TO_DATE('01-06-2024', 'DD-MM-YYYY'), 'Pending', '028', 'D20');
INSERT INTO Payment VALUES ('PM29', 'Cash', TO_DATE('25-05-2024', 'DD-MM-YYYY'), 'Received', '029', NULL);
INSERT INTO Payment VALUES ('PM30', 'COD', TO_DATE('01-06-2024', 'DD-MM-YYYY'), 'Pending', '030', 'D21');

```

```

ALTER TABLE Payment
ADD CONSTRAINT Payment_DeliveryID_FK
FOREIGN KEY (DeliveryID) REFERENCES Delivery(DeliveryID);

```

CODE 9: PAYMENTMETHODDETAIL

```

CREATE TABLE PaymentMethodDetail (
    PaymentMethod VARCHAR2(10),
    AccountUsed VARCHAR2(15) NOT NULL,
    CONSTRAINT PaymentMethodDetail_PK PRIMARY KEY (PaymentMethod),
    CONSTRAINT PaymentMethodDetail_Method_CHK CHECK (PaymentMethod IN ('Advance', 'COD', 'Cash')),
    CONSTRAINT PaymentMethodDetail_Account_CHK CHECK (AccountUsed IN ('Account A', 'Account B'))
);

INSERT INTO PaymentMethodDetail VALUES ('Advance', 'Account A');
INSERT INTO PaymentMethodDetail VALUES ('COD', 'Account B');
INSERT INTO PaymentMethodDetail VALUES ('Cash', 'Account B');

```


CODE 10: DELIVERY

```

CREATE TABLE Delivery (
    DeliveryID VARCHAR2(5),
    TrackingNumber VARCHAR2(20) NOT NULL,
    DeliveryStatus VARCHAR2(10) NOT NULL,
    DeliveryDate DATE,
    OrderID VARCHAR2(5) NOT NULL,
    CONSTRAINT Delivery_DeliveryID_PK PRIMARY KEY (DeliveryID),
    CONSTRAINT Delivery_OrderID_UQ UNIQUE (OrderID),
    CONSTRAINT Delivery_OrderID_FK FOREIGN KEY (OrderID) REFERENCES JOrder(OrderID),
    CONSTRAINT Delivery_Status_CHK CHECK (DeliveryStatus IN ('Delivered', 'Pending'))
);

INSERT INTO Delivery VALUES ('D01', 'TCS-2024-05001', 'Delivered', TO_DATE('08-05-2024','DD-MM-YYYY'), '001');
INSERT INTO Delivery VALUES ('D02', 'TCS-2024-05002', 'Delivered', TO_DATE('08-05-2024','DD-MM-YYYY'), '002');
INSERT INTO Delivery VALUES ('D03', 'TCS-2024-05003', 'Delivered', TO_DATE('09-05-2024','DD-MM-YYYY'), '003');
INSERT INTO Delivery VALUES ('D04', 'TCS-2024-05004', 'Delivered', TO_DATE('09-05-2024','DD-MM-YYYY'), '004');
INSERT INTO Delivery VALUES ('D05', 'TCS-2024-05005', 'Delivered', TO_DATE('09-05-2024','DD-MM-YYYY'), '006');
INSERT INTO Delivery VALUES ('D06', 'TCS-2024-05006', 'Delivered', TO_DATE('10-05-2024','DD-MM-YYYY'), '007');
INSERT INTO Delivery VALUES ('D07', 'TCS-2024-05007', 'Delivered', TO_DATE('14-05-2024','DD-MM-YYYY'), '008');
INSERT INTO Delivery VALUES ('D08', 'TCS-2024-05008', 'Delivered', TO_DATE('14-05-2024','DD-MM-YYYY'), '009');
INSERT INTO Delivery VALUES ('D09', 'TCS-2024-05009', 'Delivered', TO_DATE('14-05-2024','DD-MM-YYYY'), '011');
INSERT INTO Delivery VALUES ('D10', 'TCS-2024-05010', 'Delivered', TO_DATE('15-05-2024','DD-MM-YYYY'), '012');
INSERT INTO Delivery VALUES ('D11', 'TCS-2024-05011', 'Delivered', TO_DATE('19-05-2024','DD-MM-YYYY'), '014');
INSERT INTO Delivery VALUES ('D12', 'TCS-2024-05012', 'Delivered', TO_DATE('19-05-2024','DD-MM-YYYY'), '015');
INSERT INTO Delivery VALUES ('D13', 'TCS-2024-05013', 'Pending', NULL, '018');
INSERT INTO Delivery VALUES ('D14', 'TCS-2024-05014', 'Delivered', TO_DATE('20-05-2024','DD-MM-YYYY'), '020');
INSERT INTO Delivery VALUES ('D15', 'TCS-2024-05015', 'Pending', NULL, '021');
INSERT INTO Delivery VALUES ('D16', 'TCS-2024-05016', 'Pending', NULL, '022');
INSERT INTO Delivery VALUES ('D17', 'TCS-2024-05017', 'Pending', NULL, '023');
INSERT INTO Delivery VALUES ('D18', 'TCS-2024-05018', 'Pending', NULL, '026');
INSERT INTO Delivery VALUES ('D19', 'TCS-2024-05019', 'Pending', NULL, '027');
INSERT INTO Delivery VALUES ('D20', 'TCS-2024-05020', 'Pending', NULL, '028');

```

CODE 11: DEFECT

```

CREATE TABLE Defect (
    DefectID VARCHAR2(5),
    DefectDescription VARCHAR2(255) NOT NULL,
    DefectDate DATE NOT NULL,
    OrderID VARCHAR2(5) NOT NULL,
    ProductID VARCHAR2(5) NOT NULL,
    CONSTRAINT Defect_DefectID_PK PRIMARY KEY (DefectID),
    CONSTRAINT Defect_OrderID_FK FOREIGN KEY (OrderID) REFERENCES JOrder(OrderID),
    CONSTRAINT Defect_ProductID_FK FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
);

INSERT INTO Defect VALUES ('DF01', 'Wrong size delivered - ordered size 8, received size 10',
    TO_DATE('11-05-2024','DD-MM-YYYY'), '004', 'P09');
INSERT INTO Defect VALUES ('DF02', 'Fabric torn at seam on Khaadi Embroidered 3PC Khaddar',
    TO_DATE('12-05-2024','DD-MM-YYYY'), '006', 'P06');
INSERT INTO Defect VALUES ('DF03', 'Missing piece - 3PC suit had only 2 pieces in parcel',
    TO_DATE('21-05-2024','DD-MM-YYYY'), '015', 'P03');
INSERT INTO Defect VALUES ('DF04', 'Color fading - product not matching live session display',
    TO_DATE('22-05-2024','DD-MM-YYYY'), '014', 'P08');
INSERT INTO Defect VALUES ('DF05', 'Wrong item delivered - received Generation instead of Khaadi',
    TO_DATE('23-05-2024','DD-MM-YYYY'), '011', 'P06');

```

CODE 12: REFUND

```

CREATE TABLE Refund (
  RefundID VARCHAR2(5),
  RefundDate DATE NOT NULL,
  RefundAmount NUMBER(10,2) NOT NULL,
  RefundStatus VARCHAR2(10) NOT NULL,
  DefectID VARCHAR2(5) NOT NULL,
  PaymentID VARCHAR2(6) NOT NULL,
  CONSTRAINT Refund_RefundID_PK PRIMARY KEY (RefundID),
  CONSTRAINT Refund_DefectID_FK FOREIGN KEY (DefectID) REFERENCES Defect(DefectID),
  CONSTRAINT Refund_PaymentID_FK FOREIGN KEY (PaymentID) REFERENCES Payment(PaymentID),
  CONSTRAINT Refund_Amount_CHK CHECK (RefundAmount > 0),
  CONSTRAINT Refund_Status_CHK CHECK (RefundStatus IN ('Processed', 'Pending'))
);
INSERT INTO Refund VALUES ('R01', TO_DATE('14-05-2024','DD-MM-YYYY'), 3300, 'Processed', 'DF01', 'PM04');
INSERT INTO Refund VALUES ('R02', TO_DATE('15-05-2024','DD-MM-YYYY'), 7800, 'Processed', 'DF02', 'PM06');
INSERT INTO Refund VALUES ('R03', TO_DATE('24-05-2024','DD-MM-YYYY'), 3500, 'Pending', 'DF03', 'PM15');
INSERT INTO Refund VALUES ('R04', TO_DATE('26-05-2024','DD-MM-YYYY'), 4000, 'Pending', 'DF04', 'PM14');
INSERT INTO Refund VALUES ('R05', TO_DATE('28-05-2024','DD-MM-YYYY'), 7800, 'Pending', 'DF05', 'PM11');

```